TELECOM CUSTOMER

CHURN ANALYSIS

End-to-End Data Analytics Project

SQL | Python | Machine Learning | Power BI

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github.com/anjalidogra01/telecom-customer-churn-analysis

7K+	26.54%	~86%	454
Total Customers	Churn Rate	Model Accuracy	New Joiners

1. Project Overview

This project presents a complete end-to-end customer churn analysis pipeline for a telecom company. The goal is to identify customers likely to churn, understand the key drivers, and provide actionable business recommendations to reduce customer attrition.

Project Title	Telecom Customer Churn Analysis
Author	Anjali Dogra
Domain	Telecom / Customer Analytics
Tools Used	SQL Server, Python, Power BI, Excel
ML Algorithms	Random Forest with GridSearchCV Hyperparameter Tuning
Dataset	7,043 telecom customers with 38 features
GitHub	github.com/anjalidogra01/telecom-customer-churn-analysis

2. Problem Statement

Customer churn is one of the most critical challenges in the telecom industry. Acquiring a new customer costs 5-7x more than retaining an existing one. With a churn rate of 26.54%, the business is losing significant revenue. This project addresses two key questions:

- Which existing customers are at risk of churning?
- Which newly joined customers are likely to churn in the future?

A single predictive model cannot fairly answer both questions due to fundamental differences in feature distributions between new and existing customers. This project solves this through a two-model architecture.

3. Tools and Technologies

Tool	Purpose
SQL Server	ETL pipeline — null auditing, data cleaning, staging to production tables, view creation
Python	EDA, feature engineering, machine learning model building and evaluation
Sklearn	Random Forest, GridSearchCV, LabelEncoder, train-test split, classification metrics
Pandas/NumPy	Data manipulation, feature engineering, preprocessing
Matplotlib	EDA charts, confusion matrix, ROC curve, feature importance plots
Seaborn	Statistical visualizations — heatmaps, countplots, distribution plots
Power BI	Interactive dashboards — historical churn analysis and prediction results
Joblib	Model serialization — saving trained models for reuse
Excel	Prediction output file connecting Python results to Power BI dashboard

4. Data Pipeline (SQL Layer)

The data pipeline was built entirely in SQL Server following a staging-to-production pattern — a standard approach in enterprise data engineering.

4.1 Null Value Audit

A comprehensive null value check was performed across all 38 columns of the staging table using conditional SUM-CASE expressions, identifying missing values and informing the replacement strategy.

4.2 Null Value Replacement Strategy

Column Type	Replacement Logic
Service columns (Online Security, Streaming etc.)	Replaced with 'No' — NULL means service not subscribed
Internet_Type	Replaced with 'None' — NULL means no internet service
Avg Monthly Long Distance Charges	Replaced with 0 — NULL means no long distance calls
Avg Monthly GB Download	Replaced with 0 — NULL means no data usage
Churn_Category	Replaced with 'Others' — NULL means customer did not churn
Churn_Reason	Replaced with 'Others' — NULL means customer did not churn

4.3 Staging to Production

Cleaned data was inserted into a production table (prod_churn) using SELECT INTO, applying all null replacements in a single transformation step.

4.4 View Creation

Two views were created to separate customer segments:

- vw_ChurData — Churned and Stayed customers (used for model training)
- vw_JoinData — Newly joined customers (used for churn prediction)

5. Exploratory Data Analysis

EDA was performed on 7,043 customers across numerical and categorical features to understand patterns and drivers of churn before building the model.

Analysis	Finding
Customer Status Distribution	26.54% churned, 73.46% stayed
Contract vs Churn	Month-to-Month: 45.84% churn vs Two Year: 2.55%
Monthly Charge vs Churn	Churned customers have significantly higher monthly charges
Tenure vs Churn	Customers with low tenure churn more — early experience matters
Internet Type vs Churn	Fiber Optic: 40.72% churn — highest among all internet types
Payment Method vs Churn	Mailed Check: 36.88% churn — least engaged payment method
Feature Correlation	Total Revenue, Total Charges, Tenure are highly correlated

6. Model Building

6.1 Feature Engineering

Feature	Description
Tenure Group	Binned Tenure_in_Months into 5 groups: 0-1yr, 1-2yr, 2-4yr, 4-6yr, 6+yr
Revenue_Per_Month	Total_Revenue / (Tenure_in_Months + 1) — average monthly spend per customer

6.2 Model 1 — Existing Customers

Trained on historical churned and stayed customers using the full feature set including tenure, revenue, and all service usage features.

Parameter	Value
Algorithm	Random Forest Classifier
Tuning	GridSearchCV with 5-fold cross validation
Parameters tuned	n_estimators, max_depth, min_samples_split, max_features
Best parameters	n_estimators: 175, max_depth: 20, min_samples_split: 2, max_features: sqrt
Accuracy	~86%
ROC-AUC Score	High — strong discrimination between churners and stayers
Class imbalance	Handled via class_weight='balanced'
Data leakage	Fixed — encoders fitted on training data only, not full dataset

6.3 Key Issue Identified — Distribution Shift

When Model 1 was applied to new joiners, it flagged 389 out of 454 customers (85%) as churners. New customers naturally have low tenure and low revenue — which the model incorrectly interpreted as churn signals.

Feature	Historical Customers	New Joiners
Tenure_in_Months	24-48 months avg	1-3 months
Total_Revenue	High accumulated value	Very low — just joined
Number_of_Referrals	Some referrals	0 — new customer
Model output	Realistic churn risk	85% flagged as churners

6.4 Model 2 — New Joiners (Two-Model Architecture)

A separate model was built using only signup-time features available on Day 1, completely removing the tenure bias.

Parameter	Value
Signup Features Used	Gender, Age, Married, Number_of_Dependents, Contract, Payment_Method, Offer, Internet_Type, Monthly_Charge, Phone_Service, Internet_Service, Paperless_Billing
Features Excluded	Tenure_in_Months, Total_Revenue, Total_Charges, Number_of_Referrals — all time-dependent features
Algorithm	Random Forest Classifier

Parameter	Value
Class imbalance	Handled via class_weight='balanced'
Output	Churn_Probability (0-100%) + Risk Tier (High/Medium/Low) + Risk_Rank

7. Model Results

7.1 Model 1 Performance

Metric	Value
Accuracy	~86%
ROC-AUC Score	High discrimination between churners and stayers
Top Features	Tenure_in_Months, Contract, Monthly_Charge, Revenue_Per_Month
Use case	Score existing customers (tenure >= 6 months) for retention campaigns

7.2 Model 2 — New Joiner Risk Distribution

Risk Tier	Count	Percentage	Action Required
High Risk	156	34.4%	Immediate retention outreach
Medium Risk	158	34.8%	Monitor and nurture
Low Risk	140	30.8%	Standard onboarding
Total	**454**	**100%**	

The distribution is realistic and well-spread — compared to the original Model 1 output that incorrectly flagged 85% as churners. Each customer receives a Risk_Rank for prioritized retention outreach.

8. Key Insights

1. Contract type is the strongest churn driver

Month-to-Month customers churn at 45.84% vs just 2.55% for Two Year contracts — an 18x difference. New joiners on Month-to-Month contracts have an average churn probability of 62%.

2. Fiber Optic customers churn the most

Despite being the premium product, Fiber Optic has the highest churn rate at 40.72%. This points to pricing dissatisfaction or service quality issues.

3. Competitor switching is the #1 reason

841 customers left due to competitor offers — the largest churn category. The business needs proactive retention offers before customers switch.

4. Early tenure is the most vulnerable period

Customers in their first 6 months have the highest churn rate (54.71%). The first experience defines long-term retention.

5. Mailed Check payment correlates with highest churn (36.88%)
Customers not on automatic payment are least engaged and most likely to churn.

6. Premium services reduce churn significantly
83.41% of churned customers did NOT have Premium Tech Support — suggesting it is a strong retention tool.

9. Business Recommendations

#	Segment	Action	Expected Impact
1	Month-to-Month new joiners	Offer discounted annual contract at signup	Reduce High Risk count by ~30%
2	High Risk Fiber Optic customers	Free Premium Tech Support for 3 months	Address dissatisfaction churn
3	Churn_Probability >= 70%	Proactive loyalty offer before they switch	Retain before competitor loss
4	Mailed Check payment customers	Incentivize switch to auto-pay	Reduce disengagement churn
5	New joiners age 60+	Dedicated customer success call	Improve early experience

10. Power BI Dashboard

Page 1 — Churn Analysis Summary

KPI Cards	Total Customers (7K), New Joiners (454), Total Churn (1.87K), Churn Rate (26.54%)
Visuals	Churn by Contract, Age Group, Tenure Group, Internet Type, Payment Method, City
Services Table	Yes/No breakdown for all 12 service features showing churn correlation
Filters	Gender slicer, Monthly Charge Range slicer
Churn Category	Competitor (841), Dissatisfaction (321), Attitude (314), Price (211), Other (182)

Page 2 — Prediction Dashboard

KPI Cards	Total New Joiners (454), High Risk (156), Medium Risk (158), Low Risk (140)
Risk Charts	Risk tier distribution, risk by age group, internet type, payment method
Probability	Churn probability distribution histogram (binned 0-100%)
Contract Chart	Avg churn probability by contract — Month-to-Month: 62%, Two Year: 7%
Customer Table	Risk_Rank, Customer_ID, City, Contract, Internet_Type, Churn_Probability, Churn_Risk
Slicers	Churn_Risk filter (High/Medium/Low), Contract type filter

11. Conclusion

This project successfully built a complete end-to-end customer churn analytics solution covering all layers of a real-world data pipeline — from raw data cleaning in SQL to machine learning in Python to business intelligence in Power BI.

The key technical contribution is the identification and resolution of distribution shift in churn prediction. By building a two-model architecture, the project delivers reliable predictions for both existing and newly joined customers — a distinction most beginner projects overlook.

Future Improvements

- Retrain Model 2 every 6 months as new joiner outcome data accumulates
- Add SHAP values for individual customer level explainability
- Build a real-time scoring API using Flask or FastAPI
- Incorporate external data — competitor pricing and market trends
- Conduct A/B testing on retention offers to validate model ROI

Full project available on GitHub:
github.com/anjalidogra01/telecom-customer-churn-analysis

SQL scripts | Python notebook | Power BI dashboard | Excel output